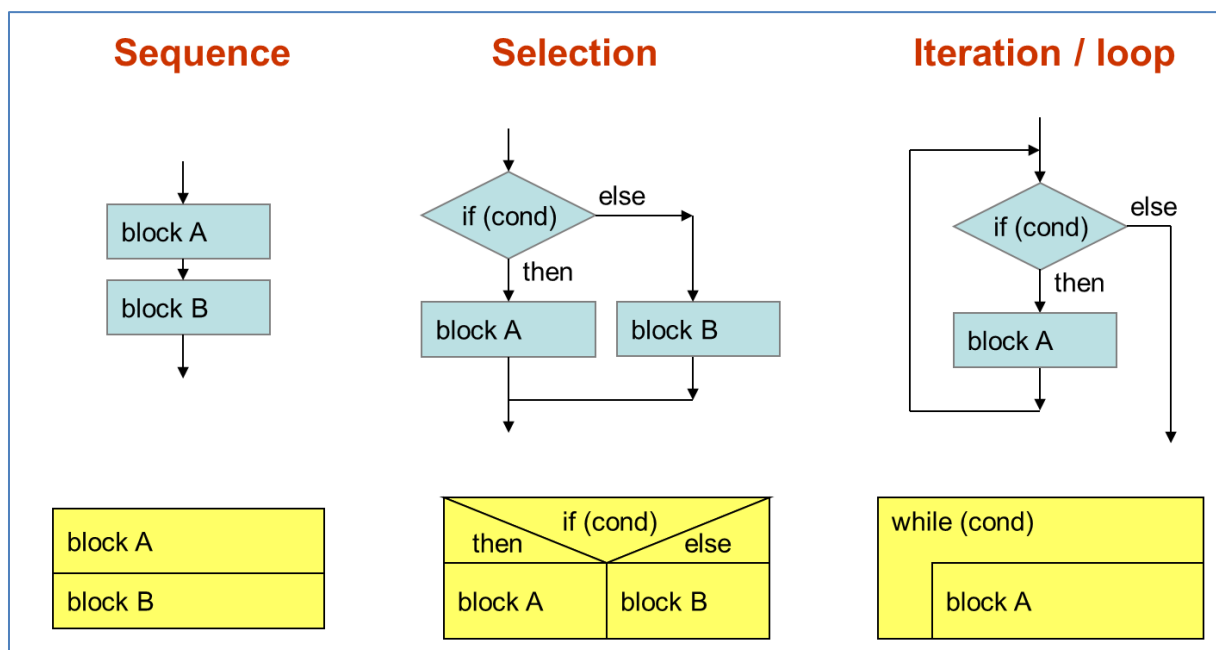


Control Structures

Content

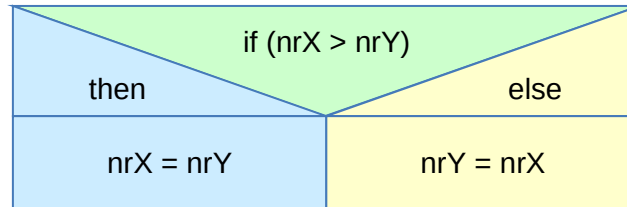
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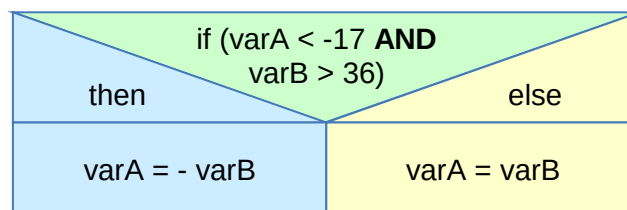
Exercise 1 – Selection/Branch

Encode the following Structograms into Flowchart, C- and ARM Assembly-language

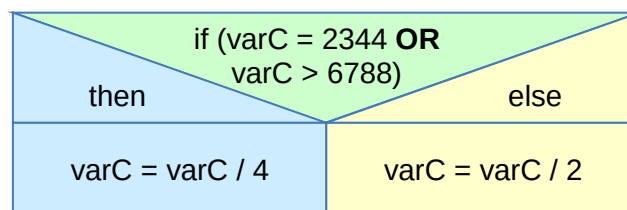
A) If-Then-Else with **unsigned** 8-bit variables



B) If-Then-Else with **signed** 8-bit variables



C) If-Then-Else with **signed** 16-bit variables



Exercise 2 – For-Loops

- A) Write a for-loop in C- and ARM Assembly-language.
- B) Compare your Assembly-language implementation with the compiler generated one.

Hint: In the Keil uVision5 IDE

- 1) create an empty C-language project (according to the respective introduction documents)
- 2) add the C-language for-loop to the empty main function
- 3) compile the project
- 4) set a breakpoint in at the first line of the main function
- 5) start debugging the program and let it run into the breakpoint
- 6) compare your Assembly-language implementation of the for-loop with the compiler generated one

Hint: for the purpose of this exercise, define your variables global and as “volatile” – this tells the compiler to not optimize away the access to the variables since they are not used otherwise.

Exercise 3 – From Code to Structogram

- A) Analyze the following Assembly-language code and derive from this the matching structogram.
- B) What result is stored in “outstr”?

```
AREA progCode, CODE, READONLY
THUMB

main PROC
EXPORT main

    LDR    R0,=srcstr
    LDR    R1,=outstr
    MOVS   R2,#0
cond     LDRB   R3,[R0,R2]
        CMP    R3,#0
        BEQ    endloop
        CMP    R3,#60
        BLO    store
        CMP    R3,#90
        BHI    store
        ADDS   R3,R3,#32
store    STRB   R3,[R1,R2]
        ADDS   R2,R2,#1
        B      cond
endloop  STRB   R3,[R1,R2]

endless B      endless
ENDP
srcstr  DCB    "This IS mY TestStriNG", 0

AREA progData, DATA, READWRITE
outstr  SPACE 50

END
```